

Continuous Functions of Two variable

①

A fn $f(x,y)$ is continuous at point (x_0, y_0) if:

- ① f is defined at (x_0, y_0)
- ② $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x,y)$ exists.
- ③ $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x,y) = f(x_0, y_0)$

A fn is continuous if it is continuous at every point of its domain.

Ex: show that

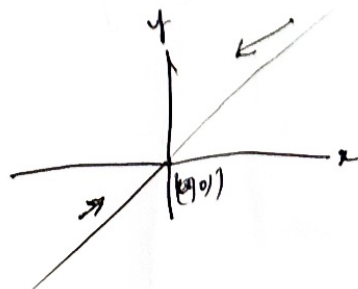
$$f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} & , (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

① is continuous at every point except the origin.

① f is defined

② $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2+y^2}$

let $y = kx$



$$\lim_{(x,y) \rightarrow (0,0)} \text{ along } y=kx$$

$$f(x,y) = \lim_{(x,y) \rightarrow (0,0)}$$

$$\left[\frac{2xy}{x^2+y^2} \right]_{y=kx}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{2x(kx)}{x^2 + (kx)^2}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{2x^2k}{x^2 + k^2x^2}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2(2k)}{x^2(1+k^2)} = \frac{2k}{1+k^2}$$

$$k=1 \quad = \frac{2}{2} = 1$$

$$k=-1 \quad = \frac{-2}{1+1} = -1$$

$y=x \rightarrow 1$
 $y=-x \rightarrow -1$

limit does not exist.

③ f_u at (x,y) , $\nexists$ limit of f_u .

Two path Test for non existence of limit (3)

If a fn $f(x,y)$ has different limits along two different paths as (x,y) approaches (x_0, y_0) then $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x,y)$ does not exist.

Continuity of composite

If f is continuous at (x_0, y_0) and g is a single variable function continuous at $f(x_0, y_0)$, then the composite function $h = g \circ f$ defined by $h(x,y) = g(f(x,y))$ is continuous at (x_0, y_0) .

For example, the composite functions

$$e^{x-y}, \quad \cos \frac{xy}{x^2+1}, \quad \ln(1+x^2y^2)$$

are continuous at every point (x,y)

Functions of more than Two variables

The definition of limit and continuity for functions of two variables and the conclusion about limits and continuity for sums, product, quotients, powers, and composites all extend to functions of three or more variables.

Extreme values of Continuous Fns on Closed, Bounded sets.

A fn of single variable that is continuous on a closed bounded interval $[a, b]$ takes on an absolute maximum value at least once in $[a, b]$.
The same is true of a fn $z = f(x, y)$ that is continuous on a closed, bounded set R in the plane (like a line segment, a disk, or a filled-in triangle).
The fn takes on an absolute maximum value at some point in R and an absolute minimum value at some point in R .

Ex 2 Find the limit

$$\lim_{P \rightarrow (1, 3, 4)} \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$$

$$\lim_{(x, y) \rightarrow (x_0, y_0)} \quad x = x_0, \quad \lim_{(x, y) \rightarrow (x_0, y_0)} y = y_0$$

$$= \lim_{P \rightarrow (1, 3, 4)} \frac{1}{x} + \lim_{P \rightarrow (1, 3, 4)} \frac{1}{y} + \lim_{P \rightarrow (1, 3, 4)} \frac{1}{z}$$

$$= \frac{1}{1} + \frac{1}{3} + \frac{1}{4} = \frac{12+4+3}{12}$$

$$= \frac{19}{12}$$

Ex: Find the limit.

⑤

$$\lim_{p \rightarrow (3,3,0)} (\sin^2 x + \cos^2 y + \sec^2 z)$$

$$= \lim_{p \rightarrow (3,3,0)} \sin^2 x + \lim_{p \rightarrow (3,3,0)} \cos^2 y + \lim_{p \rightarrow (3,3,0)} \sec^2 z$$

$$= \sin^2 3 + \cos^2 3 + 1$$

$$\sec^2 z = \frac{1}{\cos^2 z}$$

$$= 1 + 1 = 2.$$

Ex: Find the limit

$$\lim_{p \rightarrow (\pi, 0, 3)} z e^{-2y} \cos 2x$$

$$= \lim_{p \rightarrow (\pi, 0, 3)} z \cdot \lim_{p \rightarrow (\pi, 0, 3)} e^{-2y} \cdot \lim_{p \rightarrow (\pi, 0, 3)} \cos 2x$$

$$= 3 \cdot e^{-2(0)} \cdot \cos 2\pi$$

$$= 3(1)(1) = 3$$

Ex: At what points (x, y) in the plane are the function continuous?

(a) $f(x, y) = \sin(x + y)$

(b) $f(x, y) = \ln(x^2 + y^2)$

Continuity of Function of multivariables

1 at #7

In single variable we usually have the graph of single line, curve.

Geometry of continuity of one variable.

$$z = f(x, y)$$

$$\frac{dz}{dx} = \frac{d f(x, y)}{dx}$$

$$\frac{dz}{dy} = \frac{d f(x, y)}{dy}$$

Partial Differentiation / derivative

Let f be a function of x and y . If we hold y constant, say $y = y_0$ and view x as a variable, then $f(x, y_0)$ is function of x alone.

If this function is differentiable at $x = x_0$ then the value of this derivative is denoted by $f_x(x_0, y_0)$ and is called the partial derivative of f with respect of x at the point $f(x_0, y_0)$.

Similarly, if we hold x constant, say $x = x_0$ and view y as a variable, then

$f(x_0, y)$ is a func. of y -alone. if this function is differentiable at $y=y_0$, then the value of this derivative is denoted by $f_y(x_0, y_0)$ and is called partial derivative of f with respect of y at the point (x_0, y_0) .

Example

$$f(x, y) = 2x^3y^2 + 2y + 4x$$

Treating y as constant and differentiating with respect to x , we obtain

$$f_x(x, y) = 6x^2y^2 + 4$$

Treating x as constant and differentiating with respect to y , we obtain:

$$f_y(x, y) = 4x^3y + 2$$

substituting $x=1$ and $y=2$ in the partial derivative formulas yields

$$f_x(1, 2) = 6(1)^2(2)^2 + 4 = 28$$

$$f_y(1, 2) = 4(1)^3(2) + 2 = 10$$

$$\frac{\partial z}{\partial x} = 8x + 28x^3y^5$$

$$\frac{\partial z}{\partial y} = -2 + 35x^4y^4$$

Example:

$$z = f(x, y) = x^2 \sin^2 y$$

then to find the derivative of f with respect to x we treat y as a constant therefore

$$\frac{\partial z}{\partial x} = 2x \sin^2 y = f_x$$

↳ multiplied just as constant

$$\begin{aligned} \frac{\partial z}{\partial y} &= x^2 [2 \sin^2 y \cdot \cos y] \\ &= 2x^2 [\sin^2 y \cos y] \end{aligned}$$

$$\text{or } = 2x^2 \sin 2y$$

Example: $z = \ln \left[\frac{x^2 + y^2}{x + y} \right]$

$$z = \ln(x^2 + y^2) - \ln(x + y)$$

$$\frac{\partial z}{\partial x} = \frac{1}{x^2 + y^2} = \frac{1}{x^2 + y^2} \cdot 2x = \frac{2x}{x^2 + y^2}$$

$$\frac{dz}{dx} = \frac{10}{x^2 + y^2} \cdot 2x - \frac{1}{x + y}$$

$$\frac{\partial z}{\partial x} = \frac{1}{x^2+y^2} \cdot 2x - \frac{1}{x+y}$$

$$\frac{\partial z}{\partial x} = \frac{2x}{x^2+y^2} - \frac{1}{x+y}$$

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{2x^2 + 2xy - x^2 - y^2}{(x^2+y^2)(x+y)} \\ &= \frac{x^2 + 2xy - y^2}{(x^2+y^2)(x+y)} \end{aligned}$$

Similarly, (or by symmetry)

$$\frac{\partial z}{\partial y} = \frac{y^2 + 2xy - x^2}{(x^2+y^2)(x+y)}$$

Ex: $z = \cos(x^5 y^4)$

$$\begin{aligned} \frac{\partial z}{\partial x} &= -\sin(x^5 y^4) \frac{\partial}{\partial x} (x^5 y^4) \\ &= -5x^4 y^4 \sin(x^5 y^4) \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial y} &= -\sin(x^5 y^4) \frac{\partial}{\partial y} (x^5 y^4) \\ &= -\cancel{\sin y^4} = -4x^5 y^3 \sin(x^5 y^4) \end{aligned}$$

Q. 3

$$w = x^2 + 3y^2 + 4z^2 - xyz$$

(5)

$$\frac{\partial w}{\partial x} = 2x - yz$$

$$\frac{\partial w}{\partial y} = 6y - xz$$

$$\frac{\partial w}{\partial z} = 8z - xy$$